ECE-120: Basic Electronics Engineering Workshop

The Definitive University Examination Companion & Solved Papers

Complete Assessment Suite (Sets A & B for CA-1, Mid-Term, CA-2, End-Term)

Featuring Step-by-Step Mathematical Solutions & Laboratory Rubrics

- **Full CA-1 Papers (Units 1 & 2)**
- **Full Mid-Term Papers (Units 1, 2, 3)**
- **Full CA-2 Papers (Units 4 & 5)**
- **Full End-Term Papers (Units 1 to 6)**

Prepared for Lovely Professional University (LPU) Engineering Students

Course Outcomes CO1–CO6 • Bloom's Taxonomy L1–L4 • Exact Marking Rubrics

LPU Department of Electronics & Communication Engineering

Academic Session 2026–2027

Curriculum Blueprint & Examination Assessment Scheme

1. Course Articulation Matrix (Course Outcomes)

The academic syllabus of **ECE120: Basic Electronics Engineering Workshop** maps directly to six measurable Course Outcomes (CO1–CO6):

CO	Course Outcome Description	Bloom's Level
CO1	Identify passive and active electronic components, decode color codes and markings, and operate bench instrumentation (DMM, DSO, Function Generator).	L1, L2
CO2	Apply and experimentally validate the Voltage Divider Law and Current Divider Law, accounting for meter loading and burden errors.	L3, L4
CO3	Characterize the forward and reverse V-I conduction properties of Silicon PN junction diodes and evaluate rectifier circuits.	L2, L3
CO4	Verify fundamental logic gate truth tables and synthesize complex logic functions using universal NAND and NOR gates.	L3, L4
CO5	Design, wire on solderless breadboards, and evaluate combinational arithmetic circuits including Half/Full Adders and Subtractors.	L3, L4
CO6	Program, interface, and troubleshoot an Arduino Uno microcontroller with an active Infrared (IR) proximity sensor module.	L3, L4, L5

2. Examination Blueprint and Weightage Allocation

Assessment Milestones	Syllabus Coverage	Exam Format
Continuous Assessment 1 (CA-1)	Units 1 & 2	6 Subjective Problems (Attempt 5)
Mid-Term Examination	Units 1, 2, 3	10 MCQs + 4 Analytical Problems
Continuous Assessment 2 (CA-2)	Units 4 & 5	6 Subjective Problems (Attempt 5)
End-Term Examination	Units 1 to 6	Section A (MCQs) + B (Analytical) + C (Lab Desig

Quick Workshop Formula Reference Sheet

Unit 1: Components & Instruments

- Resistor: $R = (D_1 \times 10 + D_2) \times 10^M \pm \text{Tol}\,\%$
- Capacitor: $C = (D_1 \times 10 + D_2) \times 10^M$ pF
- RMS Voltage: $V_{\text{rms}} = V_m/\sqrt{2} \approx 0.707 V_m$
- Frequency: $f = 1/T$
- LED Resistor: $R_S = (V_{CC} - V_F)/I_F$

Unit 2: Divider Laws

- Voltage Divider: $V_x = V_{\text{in}}(R_x/R_{\text{total}})$
- 2-Resistor: $V_2 = V_{\text{in}} \frac{R_2}{R_1+R_2}$
- Loaded Divider: $R_{2,\text{eq}} = R_2 \parallel R_L$
- Current Divider: $I_x = I_T(G_x/G_{\text{total}})$
- 2-Branch: $I_1 = I_T \frac{R_2}{R_1+R_2}$

Unit 3: PN Junction Diode

- Shockley: $I_D = I_s(e^{V_D/\eta V_T} - 1)$
- Thermal Voltage: $V_T = kT/q \approx 26\,\text{mV}$ at 300 K
- Dynamic Resistance: $r_d = \eta V_T/I_D = 26\,\text{mV}/I_D$
- Static Resistance: $R_D = V_D/I_D$
- Half-Wave Ripple: $\gamma = 1.21$, $\eta = 40.6\%$
- Full-Wave Ripple: $\gamma = 0.482$, $\eta = 81.2\%$

Unit 4: Digital Logic & Universality

- De Morgan: $\overline{AB} = \overline{A} + \overline{B}$
- De Morgan: $\overline{A+B} = \overline{A} \cdot \overline{B}$
- XOR: $A \oplus B = \overline{A}B + A\overline{B}$
- XNOR: $\overline{A \oplus B} = AB + \overline{A}\overline{B}$
- NAND gate count: NOT=1, AND=2, OR=3, XOR=4

Unit 5: Arithmetic Circuits

- Half Adder: $S = A \oplus B$, $C = AB$
- Full Adder Sum: $S = A \oplus B \oplus C_{\text{in}}$
- Full Adder Carry: $C_{\text{out}} = AB + C_{\text{in}}(A \oplus B)$
- Half Subtractor: $D = A \oplus B$, $B_{\text{out}} = \overline{A}B$
- Full Subtractor: $D = A \oplus B \oplus B_{\text{in}}$

Unit 6: Arduino & Sensors

- ADC Voltage: $V = \text{Reading} \times (5.0/1024)\,\text{V}$
- Step size: $V_{LSB} = 4.8828\,\text{mV}$
- Maximum Pin Current: 40 mA absolute limit
- IR Comparator (LM393): Active LOW output (0 V on obstacle)

Contents

Curriculum Blueprint & Assessment Scheme **i**

Quick Workshop Formula Reference Sheet **ii**

1 Continuous Assessment 1 (CA-1) Examination Suite **1**
1.1 Continuous Assessment 1 (CA-1) — Set A 2
1.2 Continuous Assessment 1 (CA-1) — Set B 4

2 Mid-Term University Examination Suite (Units 1, 2 & 3) **6**
2.1 Mid-Term Examination — Set A 7
2.2 Mid-Term Examination — Set B 10

3 Continuous Assessment 2 (CA-2) Examination Suite **13**
3.1 Continuous Assessment 2 (CA-2) — Set A 14
3.2 Continuous Assessment 2 (CA-2) — Set B 16

4 End-Term Comprehensive University Examination Suite (Units 1 to 6) **18**
4.1 End-Term Comprehensive Examination — Set A 19
4.2 End-Term Comprehensive Examination — Set B 25

Chapter 1

Continuous Assessment 1 (CA-1) Examination Suite

Continuous Assessment 1 evaluates students on **Unit 1 (Workshop Instruments, Components, Trainer Module & Display Devices)** and **Unit 2 (Voltage and Current Divider Laws & Prototyping Measurements)**. Each paper contains **6 comprehensive questions** carrying **5 marks each**, and students attempt **5 questions** (Total = **25/30 Marks**, Time Allowed = **60 Minutes**).

1.1 Continuous Assessment 1 (CA-1) — Set A

LOVELY PROFESSIONAL UNIVERSITY

Department of Electronics & Communication Engineering

Roll No.: . . . . . **Reg. No.:** . . . . . **Section:** . . . . .

Course Code: ECE120 **Course Title:** Basic Electronics Engineering Workshop
Assessment Type: Continuous Assessment 1 (CA-1) **Set:** A
Maximum Marks: 25 Marks **Time Allowed:** 60 Minutes

Course Outcomes Evaluated: CO1 (Components & Instruments), CO2 (Divider Laws)

Instructions to Candidates: 1. Answer any **FIVE** questions out of the six questions presented. **2.** Each question carries **5 Marks**. Clearly state governing circuit theorems.

Q.1	A carbon resistor has four colored bands: Brown, Black, Orange, and Gold. Determine: (a) Nominal resistance, (b) Maximum allowable resistance, (c) Minimum allowable resistance.	**CO1**	L2	**5 Marks**
Q.2	Explain the operating principle of a Digital Multimeter when measuring resistance. Why must the circuit power be turned off before connecting DMM ohmmeter test leads?	**CO1**	L2	**5 Marks**
Q.3	A voltage divider is constructed with $V_{\text{in}} = 15\,\text{V}$, $R_1 = 3.3\,\text{k}\Omega$, and $R_2 = 6.8\,\text{k}\Omega$. Calculate: (a) The unloaded output voltage across R_2, (b) The loaded output voltage when a load resistor $R_L = 10\,\text{k}\Omega$ is connected across R_2.	**CO2**	L3	**5 Marks**
Q.4	State the Current Divider Law. For a parallel circuit with $I_T = 24\,\text{mA}$ entering two branches with $R_1 = 4\,\text{k}\Omega$ and $R_2 = 12\,\text{k}\Omega$, calculate the current in each branch.	**CO2**	L3	**5 Marks**
Q.5	A student measures a sinusoidal voltage on a DSO. The peak-to-peak waveform occupies 6 vertical divisions at $2.0\,\text{V/div}$, and one cycle spans 4 horizontal divisions at $0.5\,\text{ms/div}$. Determine: (a) Peak voltage V_m, (b) RMS voltage V_{rms}, (c) Signal frequency f.	**CO1**	L3	**5 Marks**
Q.6	A 5 mm red LED has a forward voltage drop of $V_F = 1.9\,\text{V}$ at $I_F = 15\,\text{mA}$. Design a current-limiting resistor to drive this LED safely from a $+5.0\,\text{V}$ digital logic output. Select the nearest standard 5% resistor and compute the actual LED current.	**CO1**	L4	**5 Marks**

Model Solutions & Marking Rubrics (Set A)

Step-by-Step Practical / Analytical Solution Solution to Q.1

(a) Brown = 1, Black = 0, Orange = 10^3, Gold = $\pm 5\%$.
Nominal resistance: $R = 10 \times 10^3\,\Omega = 10\,\text{k}\Omega$.
(b) Maximum limit: $R_{\max} = 10\,\text{k}\Omega + (0.05 \times 10\,\text{k}\Omega) = 10.5\,\text{k}\Omega$.
(c) Minimum limit: $R_{\min} = 10\,\text{k}\Omega - (0.05 \times 10\,\text{k}\Omega) = 9.5\,\text{k}\Omega$.

Marking Rubric (Q.1): Decoding band values: 2 Marks • Calculating nominal $10\,\text{k}\Omega$: 1 Mark • Calculating $R_{\max}$ and $R_{\min}$: 2 Marks. Total = 5 Marks.

Step-by-Step Practical / Analytical Solution Solution to Q.3

(a) Unloaded output voltage:

$$V_{\text{out,unloaded}} = V_{\text{in}} \frac{R_2}{R_1 + R_2} = 15 \times \frac{6.8}{3.3 + 6.8} = 15 \times \frac{6.8}{10.1} = 10.099\,\text{V}$$

(b) When $R_L = 10\,\text{k}\Omega$ is attached in parallel with R_2:

$$R_{2,\text{eq}} = \frac{6.8 \times 10}{6.8 + 10} = \frac{68}{16.8} = 4.048\,\text{k}\Omega$$

Loaded voltage:

$$V_{\text{out,loaded}} = 15 \times \frac{4.048}{3.3 + 4.048} = 15 \times \frac{4.048}{7.348} = 8.263\,\text{V}$$

Marking Rubric (Q.3): Correct unloaded divider formula and result ($10.10\,\text{V}$): 2 Marks • Calculation of parallel equivalent $R_2 \parallel R_L$: 1.5 Marks • Loaded output computation ($8.26\,\text{V}$): 1.5 Marks. Total = 5 Marks.

1.2 Continuous Assessment 1 (CA-1) — Set B

LOVELY PROFESSIONAL UNIVERSITY

Department of Electronics & Communication Engineering

Roll No.: . . . . . **Reg. No.:** . . . . . **Section:** . . . . .

Course Code: ECE120 **Course Title:** Basic Electronics Engineering Workshop
Assessment Type: Continuous Assessment 1 (CA-1) **Set:** B
Maximum Marks: 25 Marks **Time Allowed:** 60 Minutes

Course Outcomes Evaluated: CO1 (Components & Instruments), CO2 (Divider Laws)

Instructions to Candidates: 1. Answer any **FIVE** questions out of the six questions presented. **2.** Each question carries **5 Marks**. Clearly state governing circuit theorems.

Q.1	A precision metal-film resistor has five color bands: Red, Violet, Black, Red, Brown. Determine: (a) Nominal resistance, (b) Tolerance percentage, (c) Upper and lower tolerance limits.	**CO1**	L2	**5 Marks**
Q.2	Differentiate between an analog multimeter and a digital multimeter (DMM) in terms of input impedance, loading effect on high-resistance circuits, and measurement accuracy.	**CO1**	L2	**5 Marks**
Q.3	A $10\,\text{k}\Omega$ linear potentiometer is connected across a $+12.0\,\text{V}$ regulated supply. (a) Determine the output voltage when the wiper is positioned at 35% of its travel from ground. (b) If a load of $10\,\text{k}\Omega$ is connected to the wiper at this position, calculate the loaded voltage.	**CO2**	L3	**5 Marks**
Q.4	A total current $I_T = 50\,\text{mA}$ splits across three parallel branches with resistances $R_1 = 20\,\Omega$, $R_2 = 30\,\Omega$, and $R_3 = 60\,\Omega$. Using the conductance form of the Current Divider Law, compute the current in each branch.	**CO2**	L3	**5 Marks**
Q.5	A disc ceramic capacitor is marked with the code **473J**. (a) Decode its capacitance in picofarads, nanofarads, and microfarads. (b) State its tolerance percentage.	**CO1**	L2	**5 Marks**
Q.6	An ammeter with internal shunt resistance $R_A = 2.5\,\Omega$ is inserted to measure current in a series circuit having $V_S = 5.0\,\text{V}$ and $R = 50\,\Omega$. Calculate: (a) The true circuit current without the meter, (b) The measured circuit current, (c) The percentage insertion error.	**CO2**	L4	**5 Marks**

Model Solutions & Marking Rubrics (Set B)

Step-by-Step Practical / Analytical Solution Solution to Q.1

(a) Red = 2, Violet = 7, Black = 0, Red = $10^2 = 100$, Brown = $\pm 1\%$.
Nominal resistance: $R = 270 \times 10^2\,\Omega = 27{,}000\,\Omega = 27.0\,\text{k}\Omega$.
(b) Tolerance = $\pm 1\%$.
(c) Allowed range: $\pm 1\% \times 27\,\text{k}\Omega = \pm 0.27\,\text{k}\Omega$.
$R_{\max} = 27.27\,\text{k}\Omega$, $R_{\min} = 26.73\,\text{k}\Omega$.

Marking Rubric (Q.1): 5-band decoding: 2 Marks • Nominal value $27\,\text{k}\Omega$: 1 Mark • Tolerance bounds: 2 Marks. Total = 5 Marks.

Step-by-Step Practical / Analytical Solution Solution to Q.4

Conductances:

$$G_1 = \frac{1}{20} = 0.05\,\text{S}, \quad G_2 = \frac{1}{30} = 0.0333\,\text{S}, \quad G_3 = \frac{1}{60} = 0.0167\,\text{S}$$

Total conductance:

$$G_{\text{total}} = 0.05 + 0.0333 + 0.0167 = 0.10\,\text{S}$$

Branch currents:

$$I_1 = I_T \frac{G_1}{G_{\text{total}}} = 50 \times \frac{0.05}{0.10} = 25.0\,\text{mA}$$

$$I_2 = I_T \frac{G_2}{G_{\text{total}}} = 50 \times \frac{0.0333}{0.10} = 16.67\,\text{mA}$$

$$I_3 = I_T \frac{G_3}{G_{\text{total}}} = 50 \times \frac{0.0167}{0.10} = 8.33\,\text{mA}$$

Check: $25.0 + 16.67 + 8.33 = 50.0\,\text{mA}$ (KCL verified).

Marking Rubric (Q.4): Conductance calculations: 2 Marks • Branch current evaluations: 2.5 Marks • Verification of KCL sum: 0.5 Marks. Total = 5 Marks.

Chapter 2

Mid-Term University Examination Suite (Units 1, 2 & 3)

The Mid-Term Examination is an assessment evaluating students across **Unit 1 (Components & Test Instruments)**, **Unit 2 (Divider Laws & Loading Effects)**, and **Unit 3 (Semiconductor Diode Characteristics & Rectifiers)**. Each paper consists of **10 Objective MCQs** (1 Mark each) and **4 In-Depth Analytical/Practical Design Problems** (5 Marks each) with complete step-by-step solutions (Total = **30 Marks**, Time Allowed = **90 Minutes**).

2.1 Mid-Term Examination — Set A

LOVELY PROFESSIONAL UNIVERSITY

Department of Electronics & Communication Engineering

Roll No.: . . . . . **Reg. No.:** . . . . . **Section:** . . . . .

Course Code: ECE120 **Course Title:** Basic Electronics Engineering Workshop
Assessment Type: Mid-Term Examination **Set:** A
Maximum Marks: 30 Marks **Time Allowed:** 90 Minutes

Course Outcomes Evaluated: CO1, CO2, CO3 (Units 1, 2, 3)

Part I: Multiple-Choice Questions (10 Marks, 1 Mark Each)

Q.1 Which component opposes changes in the flow of electric current? [1 Mark]

(A) Capacitor
(B) Resistor
(C) Inductor
(D) Transformer

Q.2 A 4-band resistor with colors Yellow, Violet, Brown, Gold has nominal value: [1 Mark]

(A) 470 Ω
(B) 4.7 kΩ
(C) 47 kΩ
(D) 47 Ω

Q.3 The input impedance of a standard modern bench Digital Multimeter in DC voltage mode is: [1 Mark]

(A) 100 Ω
(B) 10 kΩ
(C) 1 MΩ
(D) 10 MΩ

Q.4 In a two-resistor voltage divider with $R_1 = R_2$, the unloaded output voltage across R_2 is: [1 Mark]

(A) V_{in}
(B) $V_{in}/2$
(C) $2V_{in}$
(D) 0 V

Q.5 A current divider consists of $R_1 = 10\,\Omega$ and $R_2 = 40\,\Omega$. The fraction of total current flowing through R_1 is: [1 Mark]

(A) 10/50
(B) $40/50 = 0.8$
(C) 50/40
(D) 1/2

Q.6 The cut-in voltage of a standard Silicon PN junction diode at room temperature is approximately: [1 Mark]

(A) 0.1 V
(B) 0.3 V
(C) 0.7 V
(D) 1.5 V

Q.7 The thermal voltage V_T at standard room temperature ($T = 300\,\text{K}$) is approximately: [1 Mark]

(A) $1.0\,\text{V}$
(B) $26\,\text{mV}$
(C) $0.7\,\text{V}$
(D) $50\,\mu\text{V}$

Q.8 The ripple factor of an unfiltered full-wave center-tapped rectifier is: [1 Mark]

(A) 1.21
(B) 0.482
(C) 0.812
(D) 0.05

Q.9 In a full-wave bridge rectifier, the Peak Inverse Voltage (PIV) rating required for each diode is: [1 Mark]

(A) $V_m/2$
(B) V_m
(C) $2V_m$
(D) $4V_m$

Q.10 When testing a healthy Silicon diode with a DMM in diode check mode in forward bias, the display reads: [1 Mark]

(A) $0.00\,\text{V}$
(B) 0.6–0.7 V
(C) OL (Overload)
(D) $5.00\,\text{V}$

Part II: Analytical & Design Problems (20 Marks, 5 Marks Each)

Q.11	A circuit has $V_{\text{in}} = 24\,\text{V}$, $R_1 = 12\,\text{k}\Omega$, and $R_2 = 12\,\text{k}\Omega$. Calculate the loading error percentage if an analog voltmeter with internal resistance $R_m = 24\,\text{k}\Omega$ is placed across R_2.	**CO2**	L3	**5 Marks**
Q.12	A Silicon PN junction diode has reverse saturation current $I_s = 5\,\text{nA}$ at $300\,\text{K}$ ($\eta = 1$). (a) Compute forward current I_D at $V_D = 0.68\,\text{V}$. (b) Compute static resistance R_D and dynamic resistance r_d at this point.	**CO3**	L3	**5 Marks**
Q.13	Design a full-wave bridge rectifier circuit fed from a $230\,\text{V}_{\text{rms}}$, $50\,\text{Hz}$ AC mains via a $10:1$ step-down transformer. Determine: (a) Peak secondary voltage V_m, (b) DC output voltage V_{dc}, (c) PIV rating per diode.	**CO3**	L4	**5 Marks**
Q.14	A sinusoidal waveform displayed on a DSO has a period of $T = 2.5\,\text{ms}$ and peak-to-peak voltage $V_{p\text{-}p} = 14\,\text{V}$. (a) Calculate signal frequency. (b) Calculate RMS voltage. (c) If horizontal timebase is $0.5\,\text{ms/div}$, how many divisions span one full cycle?	**CO1**	L3	**5 Marks**

Model Solutions & Marking Rubrics (Mid-Term Set A)

Part I: MCQ Answer Key

1: C 2: A 3: D 4: B 5: B 6: C 7: B 8: B 9: B 10: B

Step-by-Step Practical / Analytical Solution Solution to Q.12

(a) Shockley equation ($V_D/\eta V_T = 0.68/0.026 = 26.15 \gg 1$):

$$I_D = 5 \times 10^{-9} \times e^{26.15} = 5 \times 10^{-9} \times 2.276 \times 10^{11} = 1.138\,\text{A}$$

(b) Static resistance:

$$R_D = \frac{V_D}{I_D} = \frac{0.68\,\text{V}}{1.138\,\text{A}} = 0.597\,\Omega$$

Dynamic resistance:

$$r_d = \frac{\eta V_T}{I_D} = \frac{0.026\,\text{V}}{1.138\,\text{A}} = 0.0228\,\Omega$$

Marking Rubric (Q.12): Shockley calculation of I_D: 2.5 Marks • Static resistance R_D: 1 Mark • Dynamic resistance r_d: 1.5 Marks. Total = 5 Marks.

2.2 Mid-Term Examination — Set B

LOVELY PROFESSIONAL UNIVERSITY

Department of Electronics & Communication Engineering

Roll No.: .....
Reg. No.: .....
Section: .....

Course Code: ECE120 **Course Title:** Basic Electronics Engineering Workshop
Assessment Type: Mid-Term Examination **Set:** B
Maximum Marks: 30 Marks **Time Allowed:** 90 Minutes

Course Outcomes Evaluated: CO1, CO2, CO3 (Units 1, 2, 3)

Part I: Multiple-Choice Questions (10 Marks, 1 Mark Each)

Q.1 Which passive component stores energy in an electrostatic field? [1 Mark]

(A) Resistor
(B) Inductor
(C) Capacitor
(D) Thermistor

Q.2 A resistor with color bands Red, Red, Orange, Gold has nominal value: [1 Mark]

(A) 2.2 kΩ
(B) 22 kΩ
(C) 220 kΩ
(D) 22 Ω

Q.3 When an ammeter is connected into a circuit to measure branch current, it must be wired in: [1 Mark]

(A) Parallel
(B) Series
(C) Bridge
(D) Star

Q.4 The purpose of a potentiometer in an electronic workshop is: [1 Mark]

(A) Store charge
(B) Provide an adjustable voltage divider
(C) Amplify small AC signals
(D) Step down AC mains

Q.5 In a parallel circuit of two resistors R_1 and R_2, if $R_1 \ll R_2$, most of the incoming current flows through: [1 Mark]

(A) R_2
(B) R_1
(C) Equally through both
(D) Neither

Q.6 The barrier potential of a Germanium diode at room temperature is approximately: [1 Mark]

(A) 0.7 V
(B) 0.3 V
(C) 1.2 V
(D) 0.026 V

Q.7 As junction temperature increases, the reverse saturation current I_s of a diode: [1 Mark]

(A) Decreases exponentially
(B) Remains perfectly constant
(C) Doubles approximately every 10°C rise
(D) Drops to zero

Q.8 The maximum theoretical rectification efficiency of a half-wave rectifier is: [1 Mark]

(A) 40.6%
(C) 50.0%
(B) 81.2%
(D) 100%

Q.9 In a center-tapped full-wave rectifier with peak secondary winding voltage V_m across each half, the required diode PIV is: [1 Mark]

(A) V_m
(C) $4V_m$
(B) $2V_m$
(D) $V_m/2$

Q.10 A shorted diode tested with a DMM in diode check mode displays: [1 Mark]

(A) OL in both directions
(C) 0.7 V forward and OL reverse
(B) ≈ 0.00 V in both directions
(D) 10 V

Part II: Analytical & Design Problems (20 Marks, 5 Marks Each)

Q.11	A series voltage divider consists of $V_{\text{in}} = 10$ V and two 50 kΩ resistors. Calculate: (a) Unloaded output voltage across R_2, (b) Output voltage measured by a DMM with 10 MΩ input resistance, (c) Output voltage measured by an old analog meter with 100 kΩ input resistance.	**CO2**	L3	**5 Marks**
Q.12	A Silicon diode conducts 5 mA at $V_D = 0.65$ V and 20 mA at $V_D = 0.71$ V. Calculate: (a) Dynamic resistance r_d between these two operating points, (b) Static resistance at the 20 mA operating point.	**CO3**	L3	**5 Marks**
Q.13	A center-tapped full-wave rectifier delivers power to a 500 Ω load. If the AC voltage across each half of the secondary is 12 V_{rms}, calculate: (a) Peak voltage V_m, (b) DC load voltage V_{dc}, (c) Average DC load current I_{dc}, (d) Ripple voltage frequency for 50 Hz input.	**CO3**	L4	**5 Marks**
Q.14	A capacitor with code **104K** is placed in parallel with a resistor of 10 kΩ. (a) Determine the capacitance in microfarads. (b) Calculate the circuit time constant $\tau = RC$.	**CO1**	L3	**5 Marks**

Model Solutions & Marking Rubrics (Mid-Term Set B)

Part I: MCQ Answer Key

1: C 2: B 3: B 4: B 5: B 6: B 7: C 8: A 9: B 10: B

Step-by-Step Practical / Analytical Solution Solution to Q.11

(a) Unloaded output: $V_{\text{out}} = 10 \times \frac{50}{50+50} = 5.00\,\text{V}$.
(b) With DMM ($R_m = 10\,\text{M}\Omega = 10{,}000\,\text{k}\Omega$):

$$R_{2,\text{eq}} = \frac{50 \times 10000}{50 + 10000} = 49.75\,\text{k}\Omega$$

$$V_{\text{meas}} = 10 \times \frac{49.75}{50 + 49.75} = 4.987\,\text{V} \quad (\text{Error } = 0.25\%)$$

(c) With analog meter ($R_m = 100\,\text{k}\Omega$):

$$R_{2,\text{eq}} = \frac{50 \times 100}{50 + 100} = 33.33\,\text{k}\Omega$$

$$V_{\text{meas}} = 10 \times \frac{33.33}{50 + 33.33} = 4.00\,\text{V} \quad (\text{Error } = 20.0\%)$$

Marking Rubric (Q.11): Unloaded voltage calculation: 1 Mark • DMM loading calculation: 2 Marks • Analog meter severe loading calculation: 2 Marks. Total = 5 Marks.

Chapter 3

Continuous Assessment 2 (CA-2) Examination Suite

Continuous Assessment 2 evaluates students on **Unit 4 (Digital Logic Gates & Universal Gate Implementations)** and **Unit 5 (Combinational Arithmetic & Logic Circuits: Half/Full Adders & Subtractors)**. Each paper contains **6 questions** carrying **5 marks each**, with students attempting **5 questions** (Total = **25/30 Marks**, Time Allowed = **60 Minutes**).

3.1 Continuous Assessment 2 (CA-2) — Set A

LOVELY PROFESSIONAL UNIVERSITY

Department of Electronics & Communication Engineering

Roll No.: . . . . . **Reg. No.:** . . . . . **Section:** . . . . .

Course Code: ECE120 **Course Title:** Basic Electronics Engineering Workshop
Assessment Type: Continuous Assessment 2 (CA-2) **Set:** A
Maximum Marks: 25 Marks **Time Allowed:** 60 Minutes

Course Outcomes Evaluated: CO4 (Logic Gates & Universality), CO5 (Adders & Combinational)

Instructions to Candidates: 1. Answer any **FIVE** questions out of the six questions presented. **2.** Each question carries **5 Marks**. Clearly state Boolean algebra steps.

Q.1	Draw the pinout diagram of IC 7400 (Quad 2-Input NAND) showing VCC, GND, and the four gates. State the complete truth table.	**CO4**	L2	**5 Marks**
Q.2	Prove using De Morgan's theorems that a 2-input OR gate can be realized using three 2-input NAND gates. Draw the gate schematic.	**CO4**	L3	**5 Marks**
Q.3	Formulate the truth table of a Half Adder. Derive the minimal Boolean equations for Sum (S) and Carry (C). Draw the circuit using discrete XOR and AND gates.	**CO5**	L3	**5 Marks**
Q.4	Draw the logic diagram of a Full Adder constructed using two Half Adders and an OR gate. Prove algebraically that the Carry output is $C_{\text{out}} = AB + C_{\text{in}}(A \oplus B)$.	**CO5**	L4	**5 Marks**
Q.5	Synthesize a 2-input XOR gate using exactly four 2-input NAND gates. Provide the Boolean proof for the output expression.	**CO4**	L4	**5 Marks**
Q.6	Design a Half Subtractor circuit. Provide its truth table, Boolean equations for Difference (D) and Borrow (B_{out}), and breadboard implementation schematic.	**CO5**	L3	**5 Marks**

Model Solutions & Marking Rubrics (Set A)

Step-by-Step Practical / Analytical Solution Solution to Q.2

De Morgan's Theorem states: $\overline{X \cdot Y} = \overline{X} + \overline{Y}$.
To implement $Y = A + B$ using NAND gates:
1. Invert inputs A and B individually using two NAND gates configured as inverters:

$$Y_1 = \overline{A \cdot A} = \overline{A}, \quad Y_2 = \overline{B \cdot B} = \overline{B}$$

2. Feed Y_1 and Y_2 into a third NAND gate:

$$Y = \overline{Y_1 \cdot Y_2} = \overline{\overline{A} \cdot \overline{B}}$$

Applying De Morgan's theorem to the outer inversion:

$$Y = \overline{\overline{A}} + \overline{\overline{B}} = A + B$$

This realizes the OR operation using exactly 3 NAND gates.

Marking Rubric (Q.2): Statement of De Morgan's theorem: 1 Mark • Algebraic derivation: 2 Marks • Clean schematic diagram with pin connections: 2 Marks. Total = 5 Marks.

3.2 Continuous Assessment 2 (CA-2) — Set B

LOVELY PROFESSIONAL UNIVERSITY

Department of Electronics & Communication Engineering

Roll No.: . . . . . **Reg. No.:** . . . . . **Section:** . . . . .

Course Code: ECE120 **Course Title:** Basic Electronics Engineering Workshop
Assessment Type: Continuous Assessment 2 (CA-2) **Set:** B
Maximum Marks: 25 Marks **Time Allowed:** 60 Minutes

Course Outcomes Evaluated: CO4 (Logic Gates & Universality), CO5 (Adders & Combinational)

Instructions to Candidates: 1. Answer any **FIVE** questions out of the six questions presented. **2.** Each question carries **5 Marks**. Clearly state Boolean algebra steps.

Q.1	Draw the pinout diagram of IC 7402 (Quad 2-Input NOR). Highlight why its pin mapping differs from standard 7400/7408 ICs.	**CO4**	L2	**5 Marks**
Q.2	Prove algebraically that the NOR gate is universal by demonstrating the synthesis of: (a) NOT gate, (b) OR gate, (c) AND gate using only 2-input NOR gates.	**CO4**	L3	**5 Marks**
Q.3	Formulate the complete truth table of a 1-bit Full Adder with inputs A, B, C_{in} and outputs S, C_{out}. Derive the minimal sum-of-products expression for S.	**CO5**	L3	**5 Marks**
Q.4	Explain the architecture and carry propagation mechanism of the IC 7483 4-bit binary adder. What is the total delay if each stage has $t_{\text{carry}} = 10\,\text{ns}$?	**CO5**	L3	**5 Marks**
Q.5	Design a Full Subtractor circuit with inputs A, B, B_{in}. Derive minimal Boolean expressions for Difference (D) and Borrow (B_{out}).	**CO5**	L4	**5 Marks**
Q.6	What is a floating input in TTL digital circuits? Explain why floating inputs create erratic lab behavior and describe how pull-up/pull-down resistors eliminate this problem.	**CO4**	L3	**5 Marks**

Model Solutions & Marking Rubrics (Set B)

Step-by-Step Practical / Analytical Solution Solution to Q.2

(a) NOT Gate: Tie both inputs together: $Y = \overline{A + A} = \overline{A}$ (1 NOR gate).
(b) OR Gate: Invert the output of a NOR gate with a second NOR gate:

$$Y_1 = \overline{A + B}, \quad Y = \overline{Y_1 + Y_1} = \overline{\overline{A + B}} = A + B \quad \text{(2 NOR gates)}$$

(c) AND Gate: Invert inputs individually and feed into a third NOR gate:

$$Y_1 = \overline{A}, \quad Y_2 = \overline{B}$$

$$Y = \overline{Y_1 + Y_2} = \overline{\overline{A} + \overline{B}}$$

By De Morgan's theorem $\overline{X + Y} = \overline{X} \cdot \overline{Y}$:

$$Y = \overline{\overline{A}} \cdot \overline{\overline{B}} = A \cdot B \quad \text{(3 NOR gates)}$$

Marking Rubric (Q.2): NOT realization: 1 Mark • OR realization: 1.5 Marks • AND realization with De Morgan step: 2.5 Marks. Total = 5 Marks.

Chapter 4

End-Term Comprehensive University Examination Suite (Units 1 to 6)

The End-Term Examination evaluates the entire course syllabus spanning **Units 1 through 6**. Each comprehensive examination paper is structured into three sections:

- **Section A (Objective & Short Conceptual)**: 10 Questions $\times$ 2 Marks = **20 Marks**.
- **Section B (Analytical Derivations & Hardware Calculations)**: 4 Questions $\times$ 10 Marks = **40 Marks**.
- **Section C (Laboratory Experiment Scenario & Microcontroller Interfacing Design)**: 2 Comprehensive Practical System Designs $\times$ 20 Marks = **40 Marks**.

Total Marks: **100 Marks** (Scaled to official university grading weights), Time Allowed: **180 Minutes (3 Hours)**.

4.1 End-Term Comprehensive Examination — Set A

LOVELY PROFESSIONAL UNIVERSITY

Department of Electronics & Communication Engineering

Roll No.: . . . . . **Reg. No.:** . . . . . **Section:** . . . . .

Course Code: ECE120 **Course Title:** Basic Electronics Engineering Workshop
Assessment Type: End-Term Comprehensive Examination **Set:** A
Maximum Marks: 100 Marks **Time Allowed:** 180 Minutes (3 Hours)

Course Outcomes Evaluated: CO1, CO2, CO3, CO4, CO5, CO6 (Full Syllabus)

Section A: Conceptual & Short Analytical Questions (20 Marks, 2 Marks Each)

Answer all **TEN** questions:

1. What is the significance of the 4th band on a carbon resistor?
2. Explain the term 'knee voltage' of a semiconductor diode.
3. State the dual relationship between the Voltage Divider Law and the Current Divider Law.
4. Why must unused TTL logic inputs not be left floating on a breadboard?
5. State the two De Morgan's theorems.
6. Write the Boolean expression for the Sum output of a Full Adder.
7. What is the resolution in millivolts of the 10-bit ADC in an Arduino Uno with a 5.0 V reference?
8. What logic level does an active-LOW LM393 IR sensor module produce when an obstacle is detected?
9. Calculate the nominal resistance for bands: Red, Violet, Yellow, Gold.
10. What is the function of a flyback diode connected across an inductive relay coil?

Section B: Analytical Derivations & Hardware Design (40 Marks, 10 Marks Each)

Answer any **FOUR** out of five questions:

Q.11 (a) Derive the Voltage Divider Law for two series resistors R_1 and R_2 across V_{in}. (b) A 12 V divider with $R_1 = 15\,\text{k}\Omega$ and $R_2 = 15\,\text{k}\Omega$ is loaded by an external $15\,\text{k}\Omega$ resistor. Compute the loaded output voltage and the percentage loading error. **CO2**

Q.12 (a) State the Shockley diode equation and define all physical terms. (b) A Silicon diode has $I_s = 2\,\text{nA}$ at 300 K ($\eta = 1$). Calculate the diode forward current and dynamic resistance r_d at $V_D = 0.65\,\text{V}$. (c) Contrast static and dynamic resistance graphically. **CO3** L3 **10 Marks**

Q.13	(a) Prove algebraically that NAND is a universal logic gate. (b) Synthesize a 2-input XOR gate using the minimum number of 2-input NAND gates. Provide the truth table and intermediate node Boolean expressions.	**CO4**	L4	**10 Marks**
Q.14	(a) Formulate the complete truth table of a Full Adder. (b) Derive minimal Boolean expressions for Sum (S) and Carry Out (C_{out}). (c) Draw the implementation schematic using two Half Adders and an OR gate, proving algebraic equivalence.	**CO5**	L4	**10 Marks**
Q.15	Compare the performance of Half-Wave, Center-Tapped Full-Wave, and Bridge Rectifiers with respect to: (a) Number of diodes, (b) DC output voltage, (c) Ripple factor, (d) PIV rating per diode, (e) Rectification efficiency.	**CO3**	L3	**10 Marks**

Section C: Laboratory Scenario & Embedded Firmware Design (40 Marks, 20 Marks Each)

Q.16 Industrial Automated Part Counter

You are tasked with engineering a conveyor belt object counter using an Arduino Uno, an active-LOW LM393 IR sensor module, a 16×2 LCD or Serial telemetry monitor, and an alarm buzzer.

(a) Draw the complete hardware interfacing schematic showing pin connections, power supplies, and

Q.17 Workshop Breadboard Prototyping & Diagnostic Audit:

During a laboratory evaluation, a student builds a combinational arithmetic circuit on a digital trainer module. However, the circuit exhibits severe anomalies: the outputs flicker erratically, the IC becomes uncomfortably hot to the touch, and the DMM voltage reading across the +5 V rail sags to 2.8 V.

Model Solutions & Marking Rubrics (End-Term Set A)

Step-by-Step Practical / Analytical Solution Solution to Q.11

(a) By KVL in series loop: $V_{\text{in}} = I(R_1 + R_2) \implies I = \frac{V_{\text{in}}}{R_1+R_2}$.
Voltage across R_2: $V_{\text{out}} = IR_2 = V_{\text{in}}\frac{R_2}{R_1+R_2}$.
(b) Numerical calculation: Unloaded voltage: $V_{\text{unloaded}} = 12 \times \frac{15}{15+15} = 6.00\,\text{V}$.
When loaded by $R_L = 15\,\text{k}\Omega$ across R_2:

$$R_{2,\text{eq}} = 15\,\text{k}\Omega \parallel 15\,\text{k}\Omega = 7.5\,\text{k}\Omega$$

$$V_{\text{loaded}} = 12 \times \frac{7.5}{15 + 7.5} = 12 \times \frac{7.5}{22.5} = 4.00\,\text{V}$$

Loading error:

$$\%\ \text{Error} = \frac{6.00 - 4.00}{6.00} \times 100\% = \frac{2.00}{6.00} \times 100\% = 33.33\%$$

Marking Rubric (Q.11): Derivation of VDL: 4 Marks • Unloaded voltage: 2 Marks • Loaded voltage and error calculation: 4 Marks. Total = 10 Marks.

Step-by-Step Practical / Analytical Solution Solution to Q.16 (Firmware Implementation)

```
// Industrial Conveyor Object Counter with Software Debounce
const int IR_PIN = 2;       // IR sensor digital output (active LOW)
const int BUZZER_PIN = 8;   // Buzzer actuator output
const int LED_PIN = 13;     // Status LED indicator

volatile unsigned long totalCount = 0;
unsigned long lastTriggerTime = 0;
const unsigned long DEBOUNCE_TIME = 150; // ms lockout window

void setup() {
  pinMode(IR_PIN, INPUT_PULLUP);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(LED_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW);
  digitalWrite(LED_PIN, LOW);

  Serial.begin(9600);
  Serial.println("--- INDUSTRIAL CONVEYOR BATCH COUNTER ACTIVE ---");
}

void loop() {
  int sensorState = digitalRead(IR_PIN);

  if (sensorState == LOW) { // Obstacle breaks IR beam
    unsigned long now = millis();
    if (now - lastTriggerTime > DEBOUNCE_TIME) {
      totalCount++;
      lastTriggerTime = now;
```

```
      digitalWrite(LED_PIN, HIGH);

      Serial.print("Batch Counter: ");
      Serial.println(totalCount);

      // Batch alert at every 100 units
      if (totalCount % 100 == 0) {
        Serial.println("ALERT: Batch of 100 reached! Activating buzzer.");
        digitalWrite(BUZZER_PIN, HIGH);
        delay(500);
        digitalWrite(BUZZER_PIN, LOW);
      }

      delay(30);
      digitalWrite(LED_PIN, LOW);
    }
  }
}
```

Marking Rubric (Q.16): (a) Interfacing schematic: 6 Marks • (b) LM393 operational theory: 4 Marks • (c) Debounced firmware implementation with threshold condition: 10 Marks. Total = 20 Marks.

4.2 End-Term Comprehensive Examination — Set B

LOVELY PROFESSIONAL UNIVERSITY

Department of Electronics & Communication Engineering

Roll No.: . . . . . **Reg. No.:** . . . . . **Section:** . . . . .

Course Code: ECE120 **Course Title:** Basic Electronics Engineering Workshop
Assessment Type: End-Term Comprehensive Examination **Set:** B
Maximum Marks: 100 Marks **Time Allowed:** 180 Minutes (3 Hours)

Course Outcomes Evaluated: CO1, CO2, CO3, CO4, CO5, CO6 (Full Syllabus)

Section A: Conceptual & Short Analytical Questions (20 Marks, 2 Marks Each)

Answer all **TEN** questions:

1. What is the color code sequence for a $1.0\,\text{k}\Omega \pm 5\%$ resistor?
2. Define ideality factor (η) in the Shockley diode equation.
3. Why does an ammeter have low internal resistance?
4. State De Morgan's second theorem for three variables (A, B, C).
5. How many 2-input NAND gates are required to synthesize a 2-input AND gate?
6. Write the Carry Out equation for a Full Adder in terms of A, B, and C_{in}.
7. If the analog input to an Arduino Uno is 2.5 V, what is the corresponding 10-bit ADC reading?
8. State the function of the potentiometer on an IR sensor module.
9. Differentiate between dynamic resistance and static resistance of a PN junction.
10. What is the purpose of decoupling capacitors on digital circuit power rails?

Section B: Analytical Derivations & Hardware Design (40 Marks, 10 Marks Each)

Answer any **FOUR** out of five questions:

Q.11 (a) Derive the Current Divider Law for a two-branch parallel network. (b) A current source of 60 mA feeds parallel branches of $R_1 = 30\,\Omega$ and $R_2 = 60\,\Omega$. Compute the branch currents and verify Kirchhoff's Current Law. **CO2**

Q.12 (a) Explain the forward conduction characteristic of a Silicon PN diode. (b) If $I_D = 25\,\text{mA}$ at $T = 300\,\text{K}$ ($\eta = 1$), determine the dynamic AC resistance r_d. (c) Explain why the reverse saturation current increases with temperature. **CO3** L3 **10 Marks**

Q.13 (a) Prove algebraically that the NOR gate is universal. (b) Realize NOT, OR, AND, and XOR operations exclusively using 2-input NOR gates. **CO4** L4 **10 Marks**

Q.14	(a) Design a Full Subtractor circuit. Provide its truth table. (b) Derive minimal sum-of-products Boolean expressions for Difference (D) and Borrow Out (B_{out}). (c) Draw the gate-level implementation.	**CO5**	L4	**10 Marks**
Q.15	A bridge rectifier with a $1000\,\mu$F filter capacitor supplies a load of $R_L = 100\,\Omega$ from a $24\,V_{\text{rms}}, 50\,$Hz AC source. (a) Compute peak voltage V_m. (b) Calculate peak-to-peak ripple voltage $V_r = \frac{I_{dc}}{2fC}$. (c) Determine the DC load voltage.	**CO3**	L4	**10 Marks**

Section C: Laboratory Scenario & Embedded Firmware Design (40 Marks, 20 Marks Each)

Q.16 Automated Touchless Dispenser Sy

You are assigned to build a touchless liquid dispenser using an Arduino Uno, an LM393 active-LOW IR proximity sensor, a 5 V relay module, and a miniature DC pump. (a) Draw the complete wiring diagram, including the flyback protection diode across the inductive relay coil [6 Marks]. (b) Ex-

Q.17 Laboratory Measurement Accuracy & Instrument Loading Audit:

An engineering workshop team measures voltages and currents in a sensitive resistive divider network ($V_{\text{in}} = 10\,\text{V}, R_1 = 100\,\text{k}\Omega, R_2 = 100\,\text{k}\Omega$) using two different instruments: an older moving-coil analog multimeter ($20\,\text{k}\Omega/\text{V}$ on $10\,\text{V}$ range $\Longrightarrow R_m = 200\,\text{k}\Omega$) and a modern bench DMM ($R_m = 10\,\text{M}\Omega$). (a) Calculate the theo-

Model Solutions & Marking Rubrics (End-Term Set B)

Step-by-Step Practical / Analytical Solution Solution to Q.11

(a) For parallel resistors R_1 and R_2, equivalent resistance is $R_p = \frac{R_1 R_2}{R_1+R_2}$.
Voltage across parallel pair: $V_p = I_T R_p = I_T \left(\frac{R_1 R_2}{R_1+R_2}\right)$.
Current in branch 1: $I_1 = \frac{V_p}{R_1} = I_T \left(\frac{R_2}{R_1+R_2}\right)$.
Current in branch 2: $I_2 = \frac{V_p}{R_2} = I_T \left(\frac{R_1}{R_1+R_2}\right)$.
(b) Numerical calculation:

$$I_1 = 60\,\text{mA} \times \frac{60}{30+60} = 60 \times \frac{60}{90} = 40.0\,\text{mA}$$

$$I_2 = 60\,\text{mA} \times \frac{30}{30+60} = 60 \times \frac{30}{90} = 20.0\,\text{mA}$$

Sum: $I_1 + I_2 = 40.0 + 20.0 = 60.0\,\text{mA} = I_T$ (KCL verified).

Marking Rubric (Q.11): CDL algebraic derivation: 4 Marks • Branch current calculations: 4 Marks • KCL check: 2 Marks. Total = 10 Marks.

Step-by-Step Practical / Analytical Solution Solution to Q.16 (Firmware Implementation)

```
// Touchless Dispenser Controller with Hold-off Lockout
const int IR_PIN = 2;          // Active LOW IR sensor input
const int RELAY_PIN = 7;       // Relay control output pin
const int STATUS_LED = 13;     // Status LED indicator

const unsigned long DISPENSE_TIME = 1500; // 1.5 seconds pump active
const unsigned long LOCKOUT_TIME = 3000;  // 3.0 seconds cooldown period

unsigned long dispenseStartTime = 0;
unsigned long dispenseEndTime = 0;
bool isDispensing = false;

void setup() {
  pinMode(IR_PIN, INPUT_PULLUP);
  pinMode(RELAY_PIN, OUTPUT);
  pinMode(STATUS_LED, OUTPUT);
  digitalWrite(RELAY_PIN, LOW); // Relay OFF
  digitalWrite(STATUS_LED, LOW);

  Serial.begin(9600);
  Serial.println("--- TOUCHLESS DISPENSER INITIALIZED ---");
}

void loop() {
  unsigned long currentMillis = millis();
  int sensorVal = digitalRead(IR_PIN);
```

```
  if (!isDispensing) {
    // Check if cooldown lockout has elapsed
    if (currentMillis - dispenseEndTime > LOCKOUT_TIME) {
      if (sensorVal == LOW) { // Hand detected
        Serial.println("Hand Detected: Activating Dispenser Pump...");
        isDispensing = true;
        dispenseStartTime = currentMillis;
        digitalWrite(RELAY_PIN, HIGH); // Pump ON
        digitalWrite(STATUS_LED, HIGH);
      }
    }
  } else {
    // Check if dispense duration has completed
    if (currentMillis - dispenseStartTime >= DISPENSE_TIME) {
      Serial.println("Dispense Complete. Entering Cooldown Period...");
      digitalWrite(RELAY_PIN, LOW); // Pump OFF
      digitalWrite(STATUS_LED, LOW);
      isDispensing = false;
      dispenseEndTime = currentMillis;
    }
  }
}
```

Marking Rubric (Q.16): (a) Schematic with flyback diode: 6 Marks • (b) ATmega328P current limit explanation ($40\,\mathrm{mA}$ max): 4 Marks • (c) Non-blocking timing firmware with state machine logic: 10 Marks. Total = 20 Marks.